

COM S 127 - Assignment #4 Grading Rubric

Please see the 'Course Schedule' on the Syllabus for this assignment's due date.

Assignment Objective

The purpose of this assignment is to give students practice making a project from a given specification, while slowly building up their code in an iterative manner.

Instructions

Students should study the file(s) provided, and notice the various 'TODO' comments contained therein. These comments indicate places in the file that students should modify the code to be their own. These 'TODO' comments are the places in the script where code/ student work will be evaluated for the final grade on the assignment. **DO NOT PLACE CODE OUTSIDE OF THESE TODO STATEMENT AREAS.**

Students' work should be their own, completely original effort.

Files Provided

fourInSequence.py

debugging.txt

Example Output

Four In Sequence!

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[COM S 127 A]

Choice? [p]lay, [i]nstructions, [q]uit: i

The 'algorithm' of the game is thus:

- The user selects a game between one or two players.
 - A one player game has the user play against the computer.
 - A two player game has two users play against one another.
- Each player takes turns 'dropping' their 'pieces' into the board.
 - The goal is for one player to connect four pieces in an unbroken sequence.
 - A sequence can be horizontal, vertical, or diagonal.
- The game ends when one player has... FOUR IN SEQUENCE!

Choice? [p]lay, [i]nstructions, [q]uit: p

Number of Players? [1] / [2]: 1

#####

~~ Starting New Round ~~

.....
.....

.....
.....
.....
.....
0123456

Player 1, please select a column between (0-6): 0

.....
.....
.....
.....
.....
X.....
0123456

~~ End Of Player 1 (X) Turn ~~

Player 2, please select a column between (0-6): 4

.....
.....
.....
.....
.....
X...O..
0123456

~~ End Of Player 2 (O) Turn ~~

Player 1, please select a column between (0-6): 3

.....
.....
.....
.....
.....
X..XO..
0123456

~~ End Of Player 1 (X) Turn ~~

Player 2, please select a column between (0-6): 4

.....
.....
.....
.....
.....O..
X..XO..
0123456

~~ End Of Player 2 (O) Turn ~~

Player 1, please select a column between (0-6): 2

.....
.....
.....
.....
.....O..
X.XXO..
0123456

~~ End Of Player 1 (X) Turn ~~

```

Player 2, please select a column between (0-6): 1
.....
.....
.....
.....
....O..
XOXXO..
0123456

```

~~ End Of Player 2 (O) Turn ~~

```

Player 1, please select a column between (0-6): 4
.....
.....
.....
....X..
....O..
XOXXO..
0123456

```

~~ End Of Player 1 (X) Turn ~~

```

Player 2, please select a column between (0-6): 0
.....
.....
.....
....X..
O...O..
XOXXO..
0123456

```

~~ End Of Player 2 (O) Turn ~~

etc.

Special Notes

The 'algorithm' (rules) of the game is thus:

- The user selects a game between one or two players.
 - A one player game has the user play against the computer.
 - A two player game has two users play against one another.
- Each player takes turns 'dropping' their 'pieces' into the board.
 - The goal is for one player to connect four pieces in an unbroken sequence.
 - A sequence can be horizontal, vertical, or diagonal.
- The game ends when one player has... **FOUR IN SEQUENCE!**

NOTE: While there can certainly be an infinite amount of 'nitpicking' and 'nuance' applied to the algorithm above, the ultimate point is to produce a game that both works correctly, and makes sense.

- The student may have to use their own best judgement when making programming/ design decisions on this project.
- If the student's submission deviates a bit from what is demonstrated in the output above, then that

is fine. However, the student's submission should not be radically different.

NOTE: Unlike with other assignments, the `fourInSequence.py` file provided can NOT be discarded if the student wishes to make their own original version of the game. This file is the one which must be modified and submitted.

- So long as the requisite grading items are completed, then, so long as the submission works correctly, the student should receive full credit for their efforts.
- The output of the student's submission should exactly match the output seen above, as the student is not doing anything involving the script's output.

NOTE: This assignment will be more difficult than the previous assignments.

- Completing this assignment may require the student to start their work 'before the last minute.' Please plan accordingly.
- The student's script **CANNOT** crash under any circumstances. Any portions of the student's code where the script crashes will receive a zero (0) for that aspect of the script.

NOTE: The student is turning in their CODE - NOT the OUTPUT of their code.

- The student's code will be run by the TAs, and the output of those runs, along with the code itself, is what will be evaluated.

NOTE: Assignments turned in in any other format other the specified file types will not be accepted.

- Screenshots of code **will not** be accepted.
- `.sln` files are **not** code files - they contain **no** Python code and **will not** be accepted.
- `.zip`, `.rar`, `.tar.gz`, and other compressed files **will not** be accepted unless otherwise specified.
- If a student's submission is not in a `.py` file, when so specified, and if the submission is not accompanied by **all** the files needed to run the submission, the submission will not be graded.
 - **THIS WILL LEAD TO THE STUDENT RECEIVING A ZERO (0) ON THE ASSIGNMENT.**
 - **Students will NOT be allowed to re-submit their work after the final deadline in this case.**
 - Indeed, students will *not* be allowed to re-submit their work after the final deadline in *any* case.

NOTE: Some students have reported that submitting an assignment on Canvas may have difficulties associated with it.

- Submitting the assignment on some versions of Canvas may result in a 'spinning blue circle' on the submission page. This is the normal/ expected behavior.
- Students should submit their work via the GUI built into canvas - not by attempting to access the Canvas API. Doing so may cause problems.
- So long as the top of the page reads 'Submitted', and the student has re-downloaded and checked their submission, there should not be a problem.
- The student can confirm the status of their submission by downloading it themselves after they submit it. This is **highly** recommended.
 - **IF THE STUDENT DETERMINES THAT THEY HAVE TURNED IN THE**

WRONG FILE/ FILES AFTER THE FINAL DEADLINE THEY WILL *NOT* BE ALLOWED TO RE-SUBMIT THEIR WORK.

- **YOU MUST BE PROACTIVE IN YOUR OWN SUBMISSIONS!**
- **THIS *MAY* LEAD TO THE STUDENT RECEIVING A ZERO (0) ON THE ASSIGNMENT.**
 - **Students will *NOT* be allowed to re-submit their work after the final deadline in this case.**
 - Indeed, students will *not* be allowed to re-submit their work after the final deadline in *any* case.
- Students may appear to encounter difficulties in submitting their assignment due to issues with the Canvas platform itself. However, this will **not** be an excuse to turn in work past the deadline.
 - **THIS MEANS THAT STUDENTS SHOULD NOT WAIT UNTIL THE 'LAST MINUTE' TO ATTEMPT TO SUBMIT THEIR ASSIGNMENT TO CANVAS. THUS, STUDENTS WILL NEED TO 'PLAN THEIR LIVES' IN REGARD TO THEIR TIMELY SUBMISSION.**
 - **STUDENTS SHOULD ATTEMPT TO SUBMIT THEIR WORK *BEFORE* THE FINAL DEADLINE.**
 - **IF STUDENTS ENCOUNTER PROBLEMS SUBMITTING THEIR ASSIGNMENTS, PERHAPS TRY A DIFFERENT BROWSER, CLEAR THE BROWSER CACHE, ETC.**
 - **IF SUDENTS NEED HELP TURNING IN THEIR ASSIGNMENTS, THEY ARE ALWAYS WELCOME AT INSTRUCTOR/ TA OFFICE HOURS.**
 - **MANY ASSIGNMENTS HAVE BEEN SUBMITTED IN THE PAST, THUS IT *IS*, INDEED, POSSIBLE TO DO.**

NOTE: We will only grade the latest file/ files the student turns in.

- We are aware that the Canvas platform 'renames' files with a version number appended to it when subsequent submissions are made.
 - This is expected Canvas behavior.
 - This will *not* impact the portion of the grade that has to do with the filename.

Grading Items

- **(Deadline)** Was the assignment turned in by the 11:59 p.m. Friday deadline and was the file named `fourInSequence.py`?: _____ / 1
- **(Name at Top)** Has the student typed in their name/ date at the top of the `fourInSequence.py` file and deleted this 'TODO' comment?: _____ / 1
- **(Assignment #/ Description)** Did the student write the assignment number and a brief description of what the assignment is inside the initial comment block, and then delete this 'TODO' comment?: _____ / 1
- **(Name/ Section Number)** Has the student added their name/ section number to the initial script output?: _____ / 1
- **(`checkForNextMoveWin()` Function)** Has the student completed the work items specified in the `checkForNextMoveWin()` function?: _____ / 1
- **(`checkAdjacent()` Function)** Has the student completed the work items specified in the `checkAdjacent()` function?: _____ / 1
- **(`checkDraw()` Function)** Has the student completed the work items specified in the `checkDraw()` function?: _____ / 1
- **(`checkWinner()` Function)** Has the student completed the work items specified in the

checkWinner() function?: _____ / 1

- (**'i' choice tasks**) Has the student completed the work items specified by the 'i' choice tasks section?: _____ / 1
- (**'q' choice tasks**) Has the student completed the work items specified by the 'q' choice tasks section?: _____ / 1

TOTAL _____ / 10